

A critical analysis of the European Union Artificial Intelligence Act in Europe: Unveiling the Framework for Ethical AI Development in Europe.

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Abstract:

This paper critically analyses the EU AI Act in regulating ethical AI practice. The introduction already manifests the increasing influence of AI and the need for strong regulation to safeguard fundamental rights and confidence in AI systems. Besides, the thesis is further divided into four major chapters. Chapter 1 discusses manipulative AI methods, their evolution, and the steps the EU AI Act tries to undertake to prevent their harmful effects on user autonomy and the democratic process. Therefore, it explains how stakeholders struggle to create balance between innovation and the protection of fundamental rights. Chapter 2 looks at the exploitation of vulnerabilities with emphasis on age, disability, and socio-economic status. This is evaluated vis-a-vis the ethical dimensions of AI-driven personalization and the provisions of the Act in ensuring that vulnerable populations are not exploited or otherwise brought to harm. Chapter 3 criticises social scoring systems, compared with China's Social Credit System. The salient point, therefore, becomes the prohibitions within the Act against social scoring for privacy and fairness and individual freedoms. Chapter 4 elaborates on the regulation of biometric identification and real-time biometric systems in view of privacy concerns and ethical dilemmas, especially in public surveillance and law enforcement. Outlined is the way in which this act follows up on the EU legal frameworks to ensure there is coherence in the protection of the rights presented. It is arguable that, given the ambiguity, further perfection is in order, which is mostly about the enhancement of protection. The fourth part suggests ways in which the Act might be

improved to be within the EU's commitment toward responsible and ethical development of AI. In the final conclusion, this work augments the current thrust of further work toward AI governance in Europe in offering a critical analysis of the regulatory landscape.

Introduction:

Artificial intelligence should no longer be viewed as a science-fiction creation. Instead, AI should be viewed for what it is: a pervasive reality in the fabric of modern society, with implications spanning all aspects of human existence, from healthcare to banking, law enforcement to social media. This degree of pervasiveness for AI technology stems from an amazing set of dividends: decision-making efficiency, innovation, and productivity. At the same time, this glimmer of progress is fraught with ethical and regulatory obstacles. Blow up to the point where correct regulatory framework must be maintained lest the best options gain precedence over the fundamental rights decorations.

The European Union Artificial Intelligence Act (EUAI Act or AIA) is projected to cost the European economy €31 billion over the next five years. Businesses faced substantial initial compliance costs under GDPR. Large enterprises had to overhaul data management systems, train staff, and establish new protocols. Ongoing costs include maintaining compliance, handling data breaches, and potential fines¹. For SMEs, these costs were particularly burdensome, leading to concerns about competitiveness. For large enterprises, these costs might be absorbed more easily² and with only 4% of large companies against the institution it can be argued that the majority will not see the EUAI Act as a burden on their business³. Small and medium-sized enterprises (SMEs) may struggle with the financial burden, potentially stifling innovation, and competitiveness. Small and Medium-sized Enterprises

¹ “The 2022 Annual Report of the CNIL” (CNIL) <<https://www.cnil.fr/en/2022-annual-report-cnil>>.

² Deloitte, “A New Era for Privacy” (2018) <<https://www2.deloitte.com/content/dam/Deloitte/uk/Documents/risk/deloitte-uk-risk-gdpr-six-months-on.pdf>>.

³ Publications Office of the European Union, “Study to Support an Impact Assessment of Regulatory Requirements for Artificial Intelligence in Europe : Final Report.” (Publications Office of the EU, 2021) <<https://op.europa.eu/en/publication-detail/-/publication/55538b70-a638-11eb-9585-01aa75ed71a1/language-en>>.

(SMEs) deploying high-risk AI systems are expected to face compliance costs of up to €400,000⁴. The experiences of the stakeholders with the GDPR compliance provides valuable insight into the potential impact of the EUAI Act. Europe will bear the financial burden of implementing the AIA, making it crucial to discuss this topic and propose suggestions for improvement.

Importance of AI Regulation: Digital Discourse.

The European Union's AI Act represents a significant step towards establishing leadership in artificial intelligence, emphasising the importance of ethical and responsible AI development. The AI Act is crucial in setting global standards and ensuring that AI technologies serve humanity's best interests. The Act signed by all 28 Member States including Norway, underscores the collective commitment to advancing AI responsibly⁵.

By addressing potential ethical and social challenges through regulation, the AI Act aims to prevent AI from becoming a tool for exploitation and inequality. This proactive approach aligns with the broader European vision of fostering innovation while protecting fundamental rights⁶.

In the context of legitimising the status quo, myths play a crucial role. According to Gramsci (1971), dominant ideologies become powerful devices that normalise conventional wisdom into "common sense." ⁷This process of mythmaking can depoliticise discourse, masking the contested political nature of certain issues. Myths can also be used as weapons to control political debates, thereby maintaining existing power structures and reinforcing hegemonies. Common sense, as Gramsci (1971) noted, links everyday discourse with broader ideological

⁴ Ibid.

⁵ "European Approach to Artificial Intelligence" (Shaping Europe's Digital Future, April 30, 2024) <<https://digital-strategy.ec.europa.eu/en/policies/european-approach-artificial-intelligence>>.

⁶ "AI Act" (Shaping Europe's Digital Future, April 23, 2024) <<https://digital-strategy.ec.europa.eu/en/policies/regulatory-framework-ai>>.

⁷ Brevini B, 'Creating the Technological Saviour: Discourses on AI in Europe and the Legitimation of Super Capitalism' in Verdegem P (ed), AI for Everyone? Critical Perspectives (University of Westminster Press 2021) <http://www.jstor.org/stable/j.ctv26qjjhj.11>

frameworks, enriching itself with scientific ideas and philosophical opinions that permeate ordinary life. These common-sense shapes the "folklore of the future," influencing societal expectations and norms⁸. In this light, Europe's pioneering role in AI regulation is not merely about technological leadership but also about shaping the ideological landscape of AI development. By instituting comprehensive regulations, Europe aims to embed ethical considerations into the fabric of AI innovation, ensuring that the technology aligns with societal values and human rights. This approach seeks to create a future where AI contributes positively to society, reflecting the common sense and collective aspirations of its people.

Thus, the AI Act serves as a critical instrument in navigating the complex interplay between technological advancement and societal impact⁹. It positions Europe as a leader in the global discourse on AI, advocating for a balanced approach that prioritises human dignity and ethical governance. Through this legislation, Europe endeavors to set a precedent for responsible AI development worldwide, fostering an environment where AI can flourish without compromising fundamental human values¹⁰.

Chapter 1:

Chapter 1 provides a comprehensive analysis of the EU AI Act, focusing on Article 5 and its implications for AI-driven manipulative practices. It begins with an in-depth definition of Article 5, which bans AI systems using subliminal manipulative or deceptive techniques to influence behavior and impair informed decision-making, emphasising the need for comprehensive legal regulations¹¹. It highlights examples of unacceptable risks posed by algorithmic processes, particularly on social media, that shape users' behavior without

⁸ Ibid.

⁹ "European Approach to Artificial Intelligence" (Shaping Europe's Digital Future, April 30, 2024) <<https://digital-strategy.ec.europa.eu/en/policies/european-approach-artificial-intelligence>>.

¹⁰ Brevini B, 'Creating the Technological Saviour: Discourses on AI in Europe and the Legitimation of Super Capitalism' in Verdegem P (ed), *AI for Everyone? Critical Perspectives* (University of Westminster Press 2021) <http://www.jstor.org/stable/j.ctv26qjjhj.11>

¹¹ "Article 5: Prohibited Artificial Intelligence Practices | EU Artificial Intelligence Act" s Section 1 (a) <<https://artificialintelligenceact.eu/article/5/>>.

consent, interfering with privacy, freedom of expression, and democratic processes¹². The rationale behind Article 5 is examined, emphasising its role in preserving individual autonomy, transparency, and democratic processes by protecting fundamental rights, including privacy and freedom from manipulation. Criticisms of Article 5 1(a) are discussed, noting the complexity and redundancy in its language, the need for clearer definitions, and more stringent regulations to address varied manipulative tactics¹³. The document then analyses how Article 5 1(a) aligns with other EU directives and laws, such as Directive 2005/29/EC¹⁴ and Article 22 of the GDPR¹⁵, emphasising regulatory consistency and robust enforcement mechanisms. The adequacy of the European AI Office in enforcing these regulations is also discussed, highlighting challenges in keeping regulatory frameworks updated with technological advancements and calling for a more flexible, well-resourced regulatory framework. Finally, it suggests amendments to the EU AI Act, proposing that the European Artificial Intelligence Board set specific timeframes for revising regulation annexes to ensure timely updates and dynamic oversight, thus maintaining effective regulation and ethical AI development.

Chapter 2:

This chapter provides an in-depth analysis of Article 5, section 1(b) of the EU AI Act, which prohibits AI systems from exploiting vulnerabilities related to age, disability, or socioeconomic status, focusing on intentional exploitation and significant behavior distortion

¹² Nathaniel Persily and Joshua Tucker (eds), *Social Media and Democracy: The State of the Field, Prospects for Reform* (Cambridge: CUP, 2021), various chapters.

¹³ Access Now, “Access Now’s Submission to the European Commission’s Adoption Consultation on the Artificial Intelligence Act” (2021) page 13-14
<<https://www.accessnow.org/wp-content/uploads/2021/08/Submission-to-the-European-Commissions-Consultation-on-the-Artificial-Intelligence-Act.pdf>>.

¹⁴ Directive 2005/29/EC of the European Parliament and of the Council of 11 May 2005 concerning unfair business-to-consumer commercial practices in the internal market (Unfair Commercial Practices Directive) [2005] OJ L149/22.

¹⁵ Directive 2005/29/EC of the European Parliament and of the Council of 11 May 2005 concerning unfair business-to-consumer commercial practices in the internal market (Unfair Commercial Practices Directive) [2005] OJ L149/22.

causing harm¹⁶. It addresses ethical concerns about fairness, justice, and equal treatment, emphasising that targeting vulnerable groups exacerbates social disparities and fosters distrust among marginalised communities. The document highlights examples of how social media platforms use AI algorithms to personalise content feeds, exploiting human vulnerabilities like confirmation bias to boost engagement and ad revenue, which can increase societal polarisation and spread misinformation¹⁷. This practice is linked to public health concerns, such as the rise in suicide rates among young women¹⁸, underscoring the need for robust legal frameworks to protect users. The rationale behind the prohibition in Article 5 1(b) is to ensure socially just and fair use of technology, protecting those most at risk from harmful AI outcomes and promoting social equity. The document critiques the ambiguity of terms like "materially distorting behavior" and suggests that demonstrating exploitation should be sufficient for prohibition, especially for vulnerable groups like children or the disabled¹⁹. It also notes that the regulation may overlook other vulnerabilities related to gender, health, or sexual orientation. The analysis emphasises the alignment of Article 5 1(b) with GDPR²⁰ protections, highlighting how both legal frameworks work together to prevent the misuse of sensitive personal data and uphold fundamental rights to privacy, non-discrimination, and human dignity. The document concludes with a call for a

¹⁶ “Article 5: Prohibited Artificial Intelligence Practices | EU Artificial Intelligence Act” section 1(b) <<https://artificialintelligenceact.eu/article/5/>>.

¹⁷ Mendonça, Ricardo F., Fernando Filgueiras, and Virgílio Almeida, 'Algorithmic Recommenders', *Algorithmic Institutionalism: The Changing Rules of Social and Political Life* (Oxford, 2023; online edn, Oxford Academic, 14 Dec. 2023), <https://doi.org/10.1093/oso/9780192870070.003.0006>, accessed 21 May 2024.

¹⁸ Centres for Disease Control and Prevention, *Increase in Suicide Mortality in the United States, 1999-2018*, April 2020.

¹⁹ Access Now, “Access Now’s Submission to the European Commission’s Adoption Consultation on the Artificial Intelligence Act” (2021) page 14 <<https://www.accessnow.org/wp->

²⁰ Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data (General Data Protection Regulation) [2016] OJ L119/1, art 5(1)(b).

more comprehensive approach to vulnerability management to ensure protection for all individuals, regardless of their characteristics or circumstances.

Chapter 3:

Chapter 3 provides an in-depth analysis of Article 5, section 1(c) of the EU AI Act²¹, which prohibits AI systems from evaluating or classifying individuals based on social behavior or characteristics over time if it results in detrimental or unfavorable treatment. This prohibition targets unjustified actions to prevent discrimination and wrongful identification. The chapter discusses examples such as China's Social Credit System (SCS), which evaluates citizens' behaviors, leading to privileges or penalties²², and highlights its conflicts with EU principles of privacy²³ and fairness²⁴. The rationale behind the prohibition includes preventing discrimination, protecting personal freedoms, and safeguarding privacy. Critical evaluations highlight the flaws in the current regulation, such as the restriction to "over a certain period"²⁵ and the exclusion of private entities from the prohibition²⁶, which could exploit personal data without adequate oversight. The chapter also emphasises the need for sophisticated regulatory oversight to distinguish between permissible data analytics and prohibited social scoring

²¹ "Article 5: Prohibited Artificial Intelligence Practices | EU Artificial Intelligence Act" Section 1(c) <<https://artificialintelligenceact.eu/article/5/>>.

²² Amanda Lee and Amanda Lee, "What Is China's Social Credit System and Why Is It Controversial?" South China Morning Post (December 15, 2020) <<https://www.scmp.com/economy/china-economy/article/3096090/what-chinas-social-credit-system-and-why-it-controversial>>.

²³ Charter of Fundamental Rights of the European Union [2012] OJ C326/391, art 7

²⁴ Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data (General Data Protection Regulation) [2016] OJ L119/1, art 5(1)(a)

²⁵ "Article 5: Prohibited Artificial Intelligence Practices | EU Artificial Intelligence Act" Section 1(c) <<https://artificialintelligenceact.eu/article/5/>>.

²⁶ Nathalie A Smuha and others, "How the EU Can Achieve Legally Trustworthy AI: A Response to the European Commission's Proposal for an Artificial Intelligence Act" [2021] Social Science Research Network <https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3899991>.

practices. Finally, it aligns Article 5(1)(c) with GDPR provisions²⁷, ensuring lawful, fair, and transparent data processing, and underscores the importance of protecting individuals' rights and freedoms in the context of AI-driven social scoring.

Chapter 4.

Chapter 4 discusses the regulation of biometric identification and real-time remote biometric identification (RTBI) systems under the EU AI Act, highlighting significant privacy and ethical concerns due to continuous and non-consensual surveillance in public spaces. The Act proposes prohibiting RTBI systems for law enforcement unless strictly necessary and bans AI from making risk assessments solely based on profiling. It categorizes biometric systems to ensure legal compliance and addresses enforcement challenges posed by ambiguities and exemptions for law enforcement, which could lead to misuse and inadequate protection of privacy rights. Aligning with GDPR and the Law Enforcement Directive, the Act ensures lawful, fair, and transparent processing of biometric data while emphasising stringent oversight and proportionality. The chapter also discusses the impact of AI on the rule of law, noting that AI in law enforcement and public administration can undermine accountability, embed biases, and threaten the fairness of the legal system. These points underscore the need for precise regulatory measures to protect fundamental rights, prevent misuse of biometric technologies, and maintain public trust.

Chapter 5.

The European AI Office plays a crucial role in supporting the development and use of trustworthy AI, mitigating associated risks, and establishing a unified AI governance system across Europe²⁸. As AI technologies rapidly advance and the EU aims to significantly increase AI adoption among businesses, regulatory measures are essential to ensure responsible AI development and deployment. The Office, empowered by the AI Act, oversees

²⁷ Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data (General Data Protection Regulation) [2016] OJ L119/1, art 5(1)(c).

²⁸ “The AI Office: What Is It, and How Does It Work? | EU Artificial Intelligence Act” <<https://artificialintelligenceact.eu/the-ai-office-summary/>>.

AI developments, enforces regulations, and fosters international cooperation²⁹. It develops tools, methodologies, and benchmarks for evaluating AI models, provides advice and support for AI uptake, and ensures compliance with regulations. National efforts to implement the AI Act, such as those by Spain, the Netherlands, Ireland, and Luxembourg, highlight varying capacities and approaches to regulation, but concerns about effective enforcement and resource disparities persist³⁰. The European AI Office and the European AI Board are critical in providing oversight and ensuring harmonized implementation across Member States. Adequate resources and timely updates to regulations are essential to keep pace with technological advancements and address regulatory challenges.

Chapter 1: Analysis of Manipulative AI Techniques and Their Regulation under the EU AI Act

The Scope and Evolution of Manipulative AI Techniques Under Article 5 of the EU AI Act.

Article 5 of the EU AI Act bans AI systems that use subliminal, manipulative, or deceptive techniques to distort behavior and impair informed decision-making, leading individuals, or groups to make harmful decisions they wouldn't have otherwise made³¹. Manipulative or deceptive techniques in this context refer to techniques that present the information below the ‘subjective threshold’ of awareness³², combined with techniques that use deception”:

²⁹ Ibid.

³⁰ Cynthia Kroet, “As the EU AI Act Enters into Force, Focus Shifts to Countries’ Oversight Appointments” Euronews (March 12, 2024) <<https://www.euronews.com/next/2024/03/12/as-ai-act-enters-into-force-focus-shifts-to-countries-oversight-appointments>>.

³¹ “Article 5: Prohibited Artificial Intelligence Practices | EU Artificial Intelligence Act” s Section 1 (a) <<https://artificialintelligenceact.eu/article/5/>>.

³² Mary Still and Jeremiah Still, “Subliminal Techniques: Considerations and Recommendations for Analysing Feasibility,” *International Journal of Human-Computer Interaction* 35, no. 5 (2018), 457-466. See also Sid Kouider and Stanislas Dehaene “Levels of

Historically, manipulation techniques influenced people subliminally, without their conscious awareness³³. However, these methods have evolved to include more direct forms, such as chatbots designed to deceive users intentionally³⁴. Although these modern practices often bypass full conscious awareness, they are more explicit. Therefore, new legal regulations must address the full range of manipulative AI practices, covering both subtle and explicit actions. This comprehensive approach is essential to effectively protect users from all forms of AI-driven manipulation³⁵.

The Impact of Algorithmic Processes on Democratic Engagement and Privacy.

Algorithmic processes on social media have been found to act against democratic engagement by shaping users' behavior and public opinion without due disclosure³⁶. They are interfering with the privacy of people by working with collected and processed personal data without the consent of the people, leading to denying a person the right to express himself or herself, playing tricks with perception and opinion, and distorting public opinion and engagement is antithetical to improving democratic processes.

Often, these algorithms govern the flow of information and mine private data without permission, and this interferes with the most basic human right, which is the right of persons

Processing during Non-Conscious Perception: A Critical Review of Visual Masking,”
Philosophical Transactions: Biological Sciences 362, no. 1481 (2007), 857-87

³³ Nathalie A Smuha and others, “How the EU Can Achieve Legally Trustworthy AI: A Response to the European Commission’s Proposal for an Artificial Intelligence Act” [2021] Social Science Research Network 27–28
<https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3899991>.

³⁴ Benson Mawira, 'Advanced AI Models Demonstrate Ability to Deceive, Raising Ethical Concerns' (Cryptopolitan, 15 January 2024) <<https://www.cryptopolitan.com/ai-models-demonstrate-ability-to-deceive/>> accessed 17 May 2024.

³⁵ Caroline Jack, “Lexicon of Lies” (Data & Society, August 9, 2017)
<<https://datasociety.net/library/lexicon-of-lies/>>.

³⁶ Nathaniel Persily and Joshua Tucker (eds), *Social Media and Democracy: The State of the Field, Prospects for Reform* (Cambridge: CUP, 2021), various chapters.

to their privacy and freedom of expression. This may distort the public perception and engagement, a situation that is crippling to democratic processes because of the creation of echo chambers and the propagation of lies. For instance, social media may prompt mass harassment, lies, and violence alarms when the platforms boost divisive content with greater prospect of engagement through their algorithms³⁷. According to the Council of Europe and several studies, there should be regulation to safeguard democratic participation and transparency of algorithmic decision-making³⁸.

Dark patterns and subliminal manipulation both aim to deceive users and influence their behavior without their explicit awareness³⁹. Dark patterns are user interface design tricks that guide users toward actions benefiting the company⁴⁰, often by using misleading prompts or hidden costs⁴¹. Similarly, subliminal manipulation in AI involves influencing users below their level of conscious awareness, distorting their behavior and decision-making. The EU AI Act's prohibition of subliminal manipulation addresses these deceptive practices by ensuring AI systems are transparent and ethical, thereby protecting user autonomy and fundamental

³⁷ Council of Europe, “Algorithmic Processes Risk Being Used to Manipulate Social and Political Behaviours” Freedom of Expression (February 26, 2019)

<<https://www.coe.int/en/web/freedom-expression/-/algorithmic-processes-risk-being-used-to-manipulate-social-and-political-behaviours>>.

³⁸ *ibid.*

³⁹ A. M. Bhoot, M. A. Shinde, and W. P. Mishra, 'Towards the Identification of Dark Patterns: An Analysis Based on End-User Reactions' (2020) Proceedings of the 2020 CHI Conference on Human-Computer Interaction 2.

⁴⁰ Publications Office of the European Union, “Study to Support an Impact Assessment of Regulatory Requirements for Artificial Intelligence in Europe : Final Report.” (Publications Office of the EU, 2021) <<https://op.europa.eu/en/publication-detail/-/publication/55538b70-a638-11eb-9585-01aa75ed71a1/language-en>>.

⁴¹ Mathur, A., Kshirsagar, M., & Mayer, J. (2021). What makes a dark pattern... dark? design attributes, normative considerations, and measurement methods. Proceedings of the 2021 CHI Conference on Human-Computer Interaction, Section 3.1: Types of Dark Patterns.

rights⁴². This regulation combats the same kind of deceptive tactics found in dark patterns, ensuring AI technologies respect and uphold user trust and decision-making capabilities.

The Importance of Article 5,1(1) of the EU AI Act in Protecting Fundamental Rights.

Article 5 of the EU AI Act is crucial for preserving individual autonomy, transparency, and democratic processes. It safeguards fundamental rights, particularly privacy and freedom from manipulation, as outlined in the EU Charter of Fundamental Rights⁴³. The Act bans manipulative AI practices that could distort behavior or impair decision-making, thus preventing gross violations of fundamental rights such as freedom of thought, conscience, religion, and human dignity⁴⁴.

Under Article 5(1)(a), the Act protects privacy, equality, and non-discrimination. AI systems should not pose unacceptable risks to fundamental rights, which can lead to pervasive surveillance, unfair profiling, and biased decision-making. These hidden techniques violate human rights, perpetuate systemic discrimination, and harm society, which undermine trustworthy AI going against the purpose of the legislation⁴⁵. The prohibition ensures maximum legal protection, minimising harm to human dignity and promoting ethical AI development to build trust⁴⁶.

Article 5 promotes transparency and ethical AI practices, increasing trust among users and stakeholders. The EU's commitment to responsible AI development is reflected in the prohibition of AI systems that engage in subliminal manipulation. By tempering harms and preventing exploitation, the Act ensures ethical and responsible AI use.

⁴² “Liability for Damage Caused by AI Entities,” vol 11 (2022) 2 page 23-24 <<https://real-j.mtak.hu/24664/7/ACTA-LegalStudies-112-2022.pdf#page=19>>.

⁴³ “EUR-LEX - 12012P/TXT - EN - EUR-LEX” <<https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:12012P/TXT>>.

⁴⁴ “Article 5: Prohibited Artificial Intelligence Practices | EU Artificial Intelligence Act” s Section 1 (a) <<https://artificialintelligenceact.eu/article/5/>>.

⁴⁵ “Ethics Guidelines for Trustworthy AI” (Shaping Europe’s Digital Future, April 8, 2019) <<https://digital-strategy.ec.europa.eu/en/library/ethics-guidelines-trustworthy-ai>>.

⁴⁶ *ibid*.

Addressing Criticisms and Enhancing the Effectiveness of Article 5(1)(a) of the EU AI Act.

The language created under Article 5, 1(a) of the EU AI Act has faced criticism for its complexity and redundancy⁴⁷. Specifically, terms like "beyond a person's consciousness"⁴⁸ obscure rather than clarify, suggesting a need for streamlining to enhance clarity. Essentially, AI systems should be explicitly banned from using subliminal techniques that "materially distort a person's behavior."⁴⁹.

The broad regulation fails to provide clear guidance or limitations on AI use, leading to ethical concerns. The current wording lacks sufficient detail to address the varied manipulative tactics used by AI systems⁵⁰, many of which are covert but not strictly subliminal. For instance, AI models like GPT-3 have demonstrated subliminal manipulation, such as framing Muslims negatively in over 60% of cases when mentioned⁵¹. This perpetuates biases, affecting perception and decision-making, thus illustrating how AI can reinforce stereotypes.

⁴⁷ Access Now, "Access Now's Submission to the European Commission's Adoption Consultation on the Artificial Intelligence Act" (2021) page 13-14 <<https://www.accessnow.org/wp-content/uploads/2021/08/Submission-to-the-European-Commissions-Consultation-on-the-Artificial-Intelligence-Act.pdf>>.

⁴⁸ "Article 5: Prohibited Artificial Intelligence Practices | EU Artificial Intelligence Act" s Section 1 (a) <<https://artificialintelligenceact.eu/article/5/>>.

⁴⁹ *ibid*.

⁵⁰ Access Now, "Access Now's Submission to the European Commission's Adoption Consultation on the Artificial Intelligence Act" (2021) page 13-14 <<https://www.accessnow.org/wp-content/uploads/2021/08/Submission-to-the-European-Commissions-Consultation-on-the-Artificial-Intelligence-Act.pdf>>.

⁵¹ Max Tegmark and Future of Life Institute, "FLI Position Paper on the EU AI Act" (2021) page 4 <<https://futureoflife.org/wp-content/uploads/2021/08/FLI-Position-Paper-on-the-EU-AI-Act.pdf>>.

Furthermore, AI systems use emotional and political manipulation to covertly influence behaviours and attitudes⁵². This exacerbates ethical concerns, since AI uses legislative ambiguity to damage individuals and society⁵³. Without strong rules and constraints, AI will continue to push ethical boundaries, resulting in prejudice and social damage. As a result, it is important to update the EU AI Act to establish thorough and unambiguous restrictions that effectively prohibit manipulative AI techniques.

Critics argue that subjective terms like "appreciably impairing" and "purposefully manipulative"⁵⁴ create ambiguity, complicating enforcement and potentially leading to legal challenges. This vagueness makes it difficult to determine when AI practices become manipulative, leaving individuals vulnerable to covert influence that infringes on their autonomy⁵⁵. Commentators suggest that the current language undermines the prohibition's effectiveness. The EDPS-EDPB Joint Opinion also notes that the current formulation may inadequately protect fundamental rights, risking being merely symbolic and ineffective due to its limited scope.

The narrow scope of the current regulation necessitates expansion to encompass all forms of harm and interference with fundamental rights beyond physical and psychological, ensuring comprehensive protection against various manipulative technologies. While the current prohibition primarily targets subliminal techniques, it may not cover all forms of manipulation, leaving explicit manipulative practices that aren't subliminal unregulated yet harmful⁵⁶. Enforcement faces challenges, such as identifying and proving subliminal

⁵² Kate Jones and Chatham House, "AI Governance and Human Rights: Resetting the Relationship" (2023) page 31-33 <<https://www.chathamhouse.org/sites/default/files/2023-01/2023-01-10-AI-governance-human-rights-jones.pdf>>.

⁵³Future of Life Institute, 'Manipulative and Deceptive Practices by AI: Addressing Threats from the AI Act' (2022) https://futureoflife.org/wp-content/uploads/2022/08/FLI-Manipulation_AI_Act.pdf page 4-5.

⁵⁴ Article 5: Prohibited Artificial Intelligence Practices | EU Artificial Intelligence Act” s Section 1 (a) <<https://artificialintelligenceact.eu/article/5/>>.

⁵⁵ Access Now.

⁵⁶Juan Pablo Bermúdez and others, “The AI Act Needs a Practical Definition of ‘Subliminal Techniques’” www.euractiv.com (September 1, 2023)

techniques and ensuring enforcement agencies have adequate resources and expertise. Prohibiting AI systems that cause physical and psychological harm addresses the most severe impacts, preventing injury and distress. However, the regulation should broaden its definition of harm to include economic, social, and cultural harms, providing a more thorough protection framework. Currently, Article 5(1)(a) and (b) mention only physical and psychological harms, aligning with fundamental human rights but failing to address financial harms⁵⁷.

Alignment of Values and Enforcement Mechanisms in the EU AI Act and Directive 2005/29/EC

The values of the EU AI Act align closely with those under Directive 2005/29/EC, as both focus on regulatory consistency⁵⁸, protection of fundamental rights⁵⁹, and robust enforcement mechanisms⁶⁰. While Directive 2005/29/EC addresses unfair commercial practices and deceptive marketing techniques, the EU AI Act targets the subtle manipulation methods of AI systems, applying established standards to emerging technologies. Both legislative instruments aim to protect fundamental rights such as privacy, freedom of expression, and consumer rights by fostering transparency and fairness⁶¹.

<<https://www.euractiv.com/section/artificial-intelligence/opinion/the-ai-act-needs-a-practical-definition-of-subliminal-techniques/>>.

⁵⁷ “Article 5: Prohibited Artificial Intelligence Practices | EU Artificial Intelligence Act” s Section 1 (a) and section 1 (b) <<https://artificialintelligenceact.eu/article/5/>>.

⁵⁸ Directive 2005/29/EC of the European Parliament and of the Council of 11 May 2005 concerning unfair business-to-consumer commercial practices in the internal market (Unfair Commercial Practices Directive) [2005] OJ L149/22, art 3(1).

⁵⁹ Directive 2005/29/EC of the European Parliament and of the Council of 11 May 2005 concerning unfair business-to-consumer commercial practices in the internal market (Unfair Commercial Practices Directive) [2005] OJ L149/22, Recital 7.

⁶⁰ Directive 2005/29/EC of the European Parliament and of the Council of 11 May 2005 concerning unfair business-to-consumer commercial practices in the internal market (Unfair Commercial Practices Directive) [2005] OJ L149/22, arts 11 and 13.

⁶¹ Michael Veale and Frederik J Zuiderveen Borgesius, “Demystifying the Draft EU Artificial Intelligence Act — Analysing the Good, the Bad, and the Unclear Elements of the Proposed

Directive 2005/29/EC prevents consumers from unfair and misleading practices, promoting transparent and fair transactions⁶². Similarly, the EU AI Act, particularly through Article 5, protects individuals from subliminal manipulation by AI systems, ensuring autonomy, transparency, and democratic participation⁶³. Expanding the scope of the EU AI Act to include AI systems' behavior and decision-making processes would enhance its regulatory relevance and effectiveness in addressing contemporary challenges. Thus reinforcing the argument that the EUAI Act is fit for purpose.

Both directives also establish enforcement mechanisms to ensure compliance, empowering local authorities to sanction entities engaging in unfair practices, examples include Article 62 of the EUAI Act⁶⁴ and Article 11 of Directive 2005/29/EC⁶⁵ both laying the framework for enforcement. These enforcement systems increase accountability and deter legal entities from violating regulatory obligations, ensuring user safety. Thus, by aligning their objectives and enforcement approaches, both the EU AI Act and Directive 2005/29/EC contribute to a comprehensive legal framework that protects individuals from various forms of manipulation and promotes fair practices.

Approach” (2021) 22 Computer Law Review International 97, page 3-4

<<https://www.semanticscholar.org/reader/8b165eba2d0b9308682fdc4d775c00d1d3907a59>>.

⁶² Directive 2005/29/EC of the European Parliament and of the Council of 11 May 2005 concerning unfair business-to-consumer commercial practices in the internal market (Unfair Commercial Practices Directive) [2005] OJ L149/22, Annex I (List of Unfair Practices).

⁶³ 'Recital 23 | EU Artificial Intelligence Act' <https://artificialintelligenceact.eu/recital/23/> accessed 22 May 2024. Emphasises the importance of AI systems not undermining democratic processes, including elections and referenda. AI systems must not manipulate or deceive voters, ensuring that democratic participation is not compromised by AI technologies.

⁶⁴ “Article 62: Measures for Providers and Deployers, in Particular SMEs, Including Start-Ups | EU Artificial Intelligence Act” <<https://artificialintelligenceact.eu/article/62/>>.

⁶⁵ Directive 2005/29/EC of the European Parliament and of the Council of 11 May 2005 concerning unfair business-to-consumer commercial practices in the internal market (Unfair Commercial Practices Directive) [2005] OJ L149/22, art 11.

Alignment of the EU AI Act and GDPR in Protecting Against Automated Decision-Making and Manipulation.

Article 5, 1(a) of the EU AI Act⁶⁶ aligns closely with Article 22 of the GDPR⁶⁷, which addresses automated decision-making and profiling. Article 22(1) of the GDPR⁶⁸ grants individuals the right to avoid decisions based solely on automated processing that have legal or significant impacts on them. Exceptions, as outlined in Article 22(2)⁶⁹, are allowed only under specific conditions, such as explicit consent or significant contractual necessity, and must include safeguards to protect individuals' rights and interests. Similarly, Article 5, 1(a) of the EU AI Act prohibits AI systems that manipulate behavior subliminally, often involving automated decision-making without human oversight.

By banning manipulative practices, Article 5, 1(a) protects individuals from the significant negative impacts of automated decisions, ensuring that AI systems do not undermine human autonomy and decision-making, which reinforces trustworthy AI. Moreover, GDPR Article 22⁷⁰ emphasises strong safeguards for exceptions to automated decision-making, while Article 5, 1(a) of the EU AI Act sets high standards for acceptable AI practices, ensuring transparency, fairness, and regulation.

This alignment reinforces the GDPR's protective measures, creating a robust framework for safeguarding personal data and preventing misuse. Both GDPR Articles 5 and 22 emphasise transparency and fairness in data processing and automated decision-making. Article 5, 1(a) of the EU AI Act complements these principles by prohibiting manipulative AI practices,

⁶⁶ Article 5: Prohibited Artificial Intelligence Practices | EU Artificial Intelligence Act” s Section 1 (a) <<https://artificialintelligenceact.eu/article/5/>>.

⁶⁷ Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data (General Data Protection Regulation) [2016] OJ L119/1, art 22.

⁶⁸ *ibid* Article 22(1).

⁶⁹ *ibid* Article 22(2).

⁷⁰ Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data (General Data Protection Regulation) [2016] OJ L119/1, art 22.

ensuring AI systems operate ethically and legally, respecting individual rights, and promoting trust in AI technologies.

The GDPR's protection against automated decision-making is reinforced by Article 5, 1(a) of the EU AI Act, which bans subliminal manipulation by AI systems. This alignment protects human autonomy, inhibits data exploitation, and encourages openness, resulting in a strong legal framework for all parties. It protects people's personal rights and liberties. For enterprises, it offers clear criteria for ethical AI development, increasing customer trust. A unified and enforceable set of rules helps regulators better supervise and control AI technology⁷¹. Overall, this paradigm promotes the ethical progress of AI while safeguarding fundamental rights and interests across society.

Addressing Regulatory Challenges in Keeping Pace with AI Advancements.

Regulatory challenges in updating laws to keep pace with technological advancements are well-documented⁷², particularly in areas like arms control⁷³. Similar struggles are evident in AI regulation, especially concerning the prohibition of subliminal manipulation under Article 5 of the EU AI Act⁷⁴. Subliminal manipulation is a subtle, evolving threat requiring dynamic regulatory measures. The rapid evolution of AI technologies complicates this further⁷⁵, necessitating sophisticated monitoring and enforcement mechanisms. The European AI Office, responsible for enforcing these regulations, may face resource, expertise, and coordination limitations, hindering its ability to effectively address the risks posed by

⁷¹ “The Malicious Use of Artificial Intelligence: Forecasting, Prevention, and Mitigation” (February 2018) <<https://arxiv.org/pdf/1802.07228>>.

⁷² Picker, Colin, A View from 40,000 Feet: International Law and the Invisible Hand of Technology. Cardozo Law Review, Vol. 232, p. 149, 2001, Available at SSRN: <https://ssrn.com/abstract=987524>

⁷³ Amy J Nelson, “Innovation Acceleration, Digitization, and the Arms Control Imperative” [2019] Social Science Research Network <<https://doi.org/10.2139/ssrn.3382956>>.

⁷⁴ Article 5: Prohibited Artificial Intelligence Practices | EU Artificial Intelligence Act” s Section 1 (a) <<https://artificialintelligenceact.eu/article/5/>>.

⁷⁵ Picker, Colin, A View from 40,000 Feet: International Law and the Invisible Hand of Technology. Cardozo Law Review, Vol. 232, p. 149, 2001, Available at SSRN: <https://ssrn.com/abstract=987524>

subliminal manipulation. This highlights the need for a more flexible and well-resourced regulatory framework to keep up with AI advancements.

Enhancing the European AI Act Through Empowering the European Artificial Intelligence Board.

The European AI Act can be improved by empowering the European Artificial Intelligence Board (EAIB) to set specific timeframes for revising regulation Annexes⁷⁶. This would ensure timely updates, addressing challenges where static regulations fail to keep pace with AI advancements⁷⁷. A structured review process would help the EU adapt to the fast-changing AI landscape, mitigating risks like subliminal manipulation and ensuring dynamic oversight. This approach supports robust, ethical AI technology development and deployment, maintaining effective regulation.

Chapter 2: Addressing Ethical Concerns and Exploitation of Vulnerabilities under Article 5(1)(b) of the EU AI Act.

Addressing Ethical Concerns in Article 5(1)(b) of the EU AI Act.

The EU AI Act restricts AI systems from abusing vulnerabilities based on age, handicap, or socioeconomic position, with an emphasis on purposeful exploitation and severe behavior distortion that causes harm⁷⁸. Critics have questioned the effectiveness of these safeguards, particularly Article 5 section 1(b), noting ethical issues regarding fairness, justice, and equal treatment⁷⁹. Targeting vulnerable populations exacerbates socioeconomic inequalities and

⁷⁶ Luciano Floridi, “Establishing the Rules for Building Trustworthy AI” (2019) 1 Nature Machine Intelligence 261 <<https://doi.org/10.1038/s42256-019-0055-y>>.

⁷⁷ Veale, M., & Borgesius, F. J. Z. (2021). Demystifying the Draft EU Artificial Intelligence Act. Computer Law Review International, 22(4), 97-112. - Demystifying the Draft EU Artificial Intelligence Act

⁷⁸ “Article 5: Prohibited Artificial Intelligence Practices | EU Artificial Intelligence Act” section 1(b) <<https://artificialintelligenceact.eu/article/5/>>.

⁷⁹ Kate Jones and Chatham House, “AI Governance and Human Rights: Resetting the Relationship” (2023) <<https://www.chathamhouse.org/sites/default/files/2023-01/2023-01-10-AI-governance-human-rights-jones.pdf>>.

breeds distrust among marginalized areas⁸⁰. Article 5 addresses these concerns by outlawing exploitation and encouraging fairness and equitable treatment. This assures non-discrimination and protection against exploitation, while also fostering ethical AI practices and trust, so meeting the EU AI Office's purpose.

The Ethical Imperative for Regulating AI-Driven Personalisation on Social Media and Entertainment Platforms.

Social media platforms exploit AI algorithms to personalise content feeds, tapping into human vulnerabilities like confirmation bias to boost engagement and ad revenue. This practice can increase societal polarisation and spread misinformation, highlighting the need for a robust legal framework to protect users. Disturbingly, the rise in social media use since 2009 correlates with increased suicide rates among young women, with a 70% rise for ages 15-19 and a 151% rise for ages 10-14 from 2009 to 2018⁸¹.

Manipulating content feeds covertly influences users' thoughts, violating their autonomy and Article 10 of the EU Charter of Fundamental Rights⁸². The link between increased social media use and rising suicide rates among young women highlights a public health concern⁸³. Promoting addictive content harms mental health, violating the right to health and well-being. This underscores the need for a comprehensive legal framework to prevent AI from exploiting vulnerabilities and generating harmful content, protecting citizens' mental health and autonomy.

AI algorithms in social media, video platforms, and video games significantly contribute to addiction and opinion manipulation. Hyper-nudging refers to using AI algorithms and design

⁸⁰ Access Now, "Access Now's Submission to the European Commission's Adoption Consultation on the Artificial Intelligence Act" (2021) page 14
<<https://www.accessnow.org/wp-content/uploads/2021/08/Submission-to-the-European-Commissions-Consultation-on-the-Artificial-Intelligence-Act.pdf>>.

⁸¹ Centres for Disease Control and Prevention, Increase in Suicide Mortality in the United States, 1999-2018, April 2020.

⁸² EU Charter of Fundamental Rights Article 10.

⁸³ "Products - Data Briefs - Number 361 - March 2020"

<<https://www.cdc.gov/nchs/products/databriefs/db362.htm>> Centres for Disease Control and Prevention, Increase in Suicide Mortality in the United States, 1999-2018, April 2020.

techniques to direct and control user attention, often influencing behavior or opinions⁸⁴. This includes personalized recommendations on social media, targeted advertising, and content curation algorithms. While these practices may be lawful, they can also make users vulnerable to manipulation and exploitation. For instance, platforms like TikTok and Netflix use AI-powered recommendation systems to create echo chambers⁸⁵ and influence user behavior⁸⁶, reinforcing specific viewpoints and increasing user engagement to an addictive level.

By regulating these practices, the EU AI Act seeks to ensure that AI systems are used ethically and do not undermine user autonomy or well-being. This regulation is essential in curbing the potential harms of addiction and opinion manipulation fostered by advanced AI technologies.

Protecting Vulnerable Populations: The AI Act's Commitment to Social Justice and Non-Discrimination.

The EU AI Act's Article 5, 1(b) prohibits AI technologies from exploiting vulnerable groups, such as the elderly, disabled, and low socioeconomic status individuals, who lack resources to protect themselves⁸⁷. This measure ensures a socially just and fair use of technology, aligning with the Act's overarching aims of social justice and equity. It specifically protects those most

⁸⁴ Publications Office of the European Union, "Study to Support an Impact Assessment of Regulatory Requirements for Artificial Intelligence in Europe : Final Report." (Publications Office of the EU, 2021) page 31-32 <<https://op.europa.eu/en/publication-detail/-/publication/55538b70-a638-11eb-9585-01aa75ed71a1/language-en>>.

⁸⁵ Mendonça, Ricardo F., Fernando Filgueiras, and Virgílio Almeida, 'Algorithmic Recommenders', *Algorithmic Institutionalism: The Changing Rules of Social and Political Life* (Oxford, 2023; online edn, Oxford Academic, 14 Dec. 2023), <https://doi.org/10.1093/oso/9780192870070.003.0006>, accessed 21 May 2024.

⁸⁶ Stephan Manuel Nagel LIM, "Prohibited Practices under the Draft AI Act – Does the European Commission Want to Ban Instagram?" Taylor Wessing (May 21, 2024) <<https://www.taylorwessing.com/en/interface/2021/ai-act/prohibited-practices-under-the-draft-ai-act>>.

⁸⁷ "Article 5: Prohibited Artificial Intelligence Practices | EU Artificial Intelligence Act" section 1(b) <<https://artificialintelligenceact.eu/article/5/>>.

at risk from the potentially harmful outcomes of advanced AI systems. This prevents AI from worsening inequalities, ensuring technology benefits all, not just the privileged, aligning with fairness and justice to promote social equity.

The prohibition prevents discriminatory practices and manipulation based on age, disability, or socioeconomic status. Without ethical guidelines, trust in AI systems erodes, leading to misuse and diminished public confidence⁸⁸. This undermines innovation, as stakeholders less likely to trust AI advancements may hesitate to adopt the technology.

By targeting protections towards vulnerable groups, Article 5, 1(b) reinforces anti-discrimination laws and human rights frameworks. It ensures that AI deployment does not lead to further marginalization of these groups, thereby upholding the right to non-discrimination and equality for all individuals. This is particularly important in preventing AI systems from profiling and discriminating against individuals based on their physical characteristics or other personal attributes.

Addressing Ambiguities and Expanding Protections in Article 5, Paragraph 1(b) of the EU AI Act.

The ambiguity of "materially distorting behavior" in Article 5, paragraph 1(b) of the EU AI Act is problematic⁸⁹. Requiring AI to exploit vulnerabilities and cause harm sets a high threshold. Critics argue that distorting behavior already infringes on rights, making the harm requirement redundant⁹⁰. Exploiting vulnerabilities should be sufficient for prohibition, especially for groups like children or the disabled. Victims shouldn't have to prove harm; demonstrating exploitation should be enough for a ban.

⁸⁸ "Ethics of Artificial Intelligence and Robotics (Stanford Encyclopedia of Philosophy)" (April 30, 2020) <<https://plato.stanford.edu/entries/ethics-ai/>>. various pages

⁸⁹ Access Now, "Access Now's Submission to the European Commission's Adoption Consultation on the Artificial Intelligence Act" (2021) page 14 <<https://www.accessnow.org/wp-content/uploads/2021/08/Submission-to-the-European-Commissions-Consultation-on-the-Artificial-Intelligence-Act.pdf>>.

⁹⁰ *ibid*.

Article 5, paragraph 1(b) of the EU AI Act may overlook vulnerabilities related to gender, health, or sexual orientation by limiting criteria to age and disability⁹¹. This narrow focus fails to address the broader spectrum of exploitable traits by AI systems. Identifying individuals within these groups can also be invasive and challenging⁹², especially with complex socioeconomic issues. A more comprehensive vulnerability management approach is needed to ensure protection for all individuals, regardless of their characteristics or circumstances.

Harmonising Data Protection: The GDPR and AI Act's Unified Approach to Safeguarding Sensitive Information and Fundamental Rights.

Article 9 of the GDPR provides stringent protections for processing special categories of personal data, such as racial or ethnic origin, political opinions, religious beliefs, trade union membership, genetic and biometric data, health data, and data concerning a person's sex life or sexual orientation⁹³. Processing such data is generally prohibited, except under specific conditions like explicit consent⁹⁴, vital interests, or substantial public interest⁹⁵.

⁹¹ “Article 5: Prohibited Artificial Intelligence Practices | EU Artificial Intelligence Act” section 1(b) <<https://artificialintelligenceact.eu/article/5/>>.

⁹² Access Now, “Access Now’s Submission to the European Commission’s Adoption Consultation on the Artificial Intelligence Act” (2021) page 14 <<https://www.accessnow.org/wp-content/uploads/2021/08/Submission-to-the-European-Commissions-Consultation-on-the-Artificial-Intelligence-Act.pdf>>.

⁹³ Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data (General Data Protection Regulation) [2016] OJ L119/1, art 9.

⁹⁴ Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data (General Data Protection Regulation) [2016] OJ L119/1, art 9(2)(a).

⁹⁵ Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data (General Data Protection Regulation) [2016] OJ L119/1, art 9(2)(g).

The AI Act's prohibition on exploiting vulnerabilities aligns closely with these GDPR protections. By preventing AI systems from targeting individuals based on age, disability, or socioeconomic status, the AI Act complements the GDPR's restrictions on processing sensitive data. This alignment ensures both legal frameworks work together to provide robust protection against the misuse of sensitive personal data.

The AI Act's prohibition helps prevent this misuse, reinforcing the GDPR's protections. This dual approach ensures AI systems cannot exploit personal data to manipulate or harm individuals, upholding fundamental rights to privacy, non-discrimination⁹⁶, and human dignity⁹⁷ as protected by both the GDPR and the AI Act.

Chapter 3: Analysis of Social Scoring AI Techniques and Their Regulation under the EU AI Act.

The Scope and Evolution of Social Scoring Under Article 5 of the EU AI Act.

Article 5, 1(c) of the EU AI Act prohibits AI systems from evaluating or classifying individuals or groups based on social behavior or characteristics over time, whether known, inferred, or predicted, if it results in detrimental or unfavorable treatment unrelated to the original context of the data collection⁹⁸. This prohibition targets unjustified or disproportionate actions and aims to prevent discrimination and wrongful identification due to biased, non-contextual inferences from behavior or characteristics. The provision ensures that AI systems do not misuse social scoring to unfairly treat individuals, upholding principles of fairness and non-discrimination⁹⁹.

⁹⁶ Charter of Fundamental Rights of the European Union [2012] OJ C326/391, art 21.

⁹⁷ Charter of Fundamental Rights of the European Union [2012] OJ C326/391, art 1.

⁹⁸ “Article 5: Prohibited Artificial Intelligence Practices | EU Artificial Intelligence Act” Section 1(c) <<https://artificialintelligenceact.eu/article/5/>>.

⁹⁹ Publications Office of the European Union, “Study to Support an Impact Assessment of Regulatory Requirements for Artificial Intelligence in Europe : Final Report.” (Publications Office of the EU, 2021) page 7-8 <<https://op.europa.eu/en/publication-detail/-/publication/55538b70-a638-11eb-9585-01aa75ed71a1/language-en>>.

A Shield Against the Privacy Infringements of China's Social Credit System and Lessons from the Gladsaxe Data Leak.

China's Social Credit System (SCS) is a state-run program evaluating citizens' and businesses' trustworthiness by collecting and analysing data on behaviours such as financial records, social conduct, and law adherence¹⁰⁰. High scores grant privileges like easy access to loans, free travel, and high-quality jobs, while low scores result in penalties such as travel restrictions, job bans, and limited access to services¹⁰¹. The system's heavy surveillance raises significant privacy concerns, as data is constantly acquired and analysed without explicit consent, compromising individuals' privacy rights.

The SCS's lack of transparency and government surveillance would violate the EU's GDPR principles, which emphasise individual consent¹⁰² and data transparency¹⁰³. The fairness of the SCS is also questionable, as judgments are based on data that often do not accurately reflect an individual's character or circumstances. This pushes individuals to conform to state-defined behavior standards, eroding personal freedom¹⁰⁴.

¹⁰⁰ Rogier Creemers, "China's Social Credit System: An Evolving Practice of Control" [2018] Social Science Research Network

<https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3175792>.

¹⁰¹ Genia Kostka, "China's Social Credit Systems and Public Opinion: Explaining High Levels of Approval" [2018] Social Science Research Network

<https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3215138>.

¹⁰² Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data (General Data Protection Regulation) [2016] OJ L119/1, art 7.

¹⁰³ Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data (General Data Protection Regulation) [2016] OJ L119/1, art 5.

¹⁰⁴ Charter of Fundamental Rights of the European Union [2012] OJ C326/391, art 10.

EU laws on discrimination, such as Directive 2000/78/EC, ensure that individuals are not discriminated against based on irrelevant grounds¹⁰⁵. The SCS categorizes individuals by their scores¹⁰⁶, potentially compromising the EU's commitment to fairness and personal autonomy. Additionally, the SCS encourages societal conformity by rewarding behaviours aligned with government standards and penalising non-conformity, repressing individual expression and dissent¹⁰⁷. It can be used as an instrument of political repression to monitor and target dissidents, activists, and other undesirables, undermining democratic principles and human rights.

The EU, founded on democratic values, human rights, and law¹⁰⁸, supports a pluralistic society with freedom of expression for all viewpoints. Using the SCS to spy on and punish political dissent contradicts the EU's core values and individual rights, making it unacceptable within the EU. Therefore, the SCS's approach fundamentally clashes with the EU's principles of privacy, fairness, and freedom. The EUAI Act prohibits the act of social scoring for these purposes. Therefore, advocating for trustworthy AI.

In Gladsaxe, Denmark, a tracing tool was implemented as part of the country's "ghetto plan" to identify vulnerable children early on¹⁰⁹. This system allowed municipalities to collect and combine data from various public sources to categorize families based on risk indicators, assigning scores based on criteria such as attendance at doctor's appointments, employment status, and mental health. However, in December 2018, a data leak exposed the personal information of over 20,000 citizens, including details on gender, age, welfare benefits, and

¹⁰⁵ Council Directive 2000/78/EC of 27 November 2000 establishing a general framework for equal treatment in employment and occupation [2000] OJ L303/16, art 2(1).

¹⁰⁶ Amanda Lee and Amanda Lee, "What Is China's Social Credit System and Why Is It Controversial?" South China Morning Post (December 15, 2020) <<https://www.scmp.com/economy/china-economy/article/3096090/what-chinas-social-credit-system-and-why-it-controversial>>.

¹⁰⁷ Darrell M West and Jack Karsten, "China's Social Credit System Spreads to More Daily Transactions" Brookings (June 18, 2018) <<https://www.brookings.edu/articles/chinas-social-credit-system-spreads-to-more-daily-transactions/>>.

¹⁰⁸ Consolidated Version of the Treaty on European Union [2012] OJ C326/13, art 2.

¹⁰⁹ "Denmark - Automating Society Report 2020" (Automating Society Report 2020) <<https://automatingsociety.algorithmwatch.org/report2020/denmark/>>

family conditions¹¹⁰. This incident highlighted significant privacy and data protection risks associated with profiling programs, as many individuals were unaware of their inclusion and had no means to object. The EU AI Act aims to mitigate such risks by regulating AI applications, ensuring they do not exploit vulnerabilities, infringe on privacy, or operate without transparency, thereby safeguarding individual rights and promoting ethical AI practices.

Rationale Behind the Prohibition of Social Scoring.

Article 41 of the EU Charter of Fundamental Rights grants individuals the right to have their affairs handled impartially, fairly, and within a reasonable time by EU institutions. Article 5 of the EU AI Act seeks to protect individuals from manipulative and non-transparent AI practices that can infringe on fundamental rights, including the right to good administration. By prohibiting AI systems that operate non-transparently and manipulate behavior without user awareness, Article 5 aims to uphold the principles of fairness, impartiality, and accountability in administrative processes. This regulation is crucial in preventing discrimination and ensuring that citizens receive adequate explanations for decisions that affect them, thereby safeguarding their fundamental rights and promoting trust in public administration.

Strengthening the EU AI Act's Ban on Social Scoring for Comprehensive Protection.

Critics argue that Article 5, paragraph 1(c) of the EU AI Act's ban on social scoring is flawed. It restricts scoring "over a certain period," ignoring immediate or cumulative scoring's potential to violate fundamental rights and privacy, thus failing to protect against significant harm¹¹¹. The current language of the regulation requires AI systems to exploit vulnerabilities "over a certain period," which can be difficult to monitor and prove. The vague definition of

¹¹⁰ "Confidential Data Loss in Gladsaxe Denmark"

<<https://www.systemtek.co.uk/2018/12/confidential-data-loss-in-gladsaxe-denmark/>>.

¹¹¹ Access Now, "Access Now's Submission to the European Commission's Adoption Consultation on the Artificial Intelligence Act" (2021) page 15-16

<<https://www.accessnow.org/wp-content/uploads/2021/08/Submission-to-the-European-Commissions-Consultation-on-the-Artificial-Intelligence-Act.pdf>>.

what constitutes "social scoring" and the specific contexts in which it is prohibited add to the complexity of implementation.

Another significant criticism is that the EU AI Act's prohibition on social scoring is limited to public authorities, creating a loophole for private entities, especially in social media and cloud services, to engage in similar practices without adequate oversight. Critics call for broader coverage to include all social scoring activities¹¹², clarify "on behalf of" public authorities, and expand criteria to include proxies like geographical location to prevent discrimination and harm¹¹³.

Private companies, especially social media and cloud service providers can process vast amounts of personal data and conduct social scoring. Without extending the prohibition to cover private entities, there is a risk that these companies will use social scoring to the detriment of individuals' privacy, autonomy, and equal treatment¹¹⁴. The European Data Protection Supervisor (EDPS) and European Data Protection Board (EDPB)¹¹⁵ have emphasised the need for stricter oversight and broader regulatory coverage to include these private actors, ensuring they do not exploit loopholes to harm individuals¹¹⁶.

¹¹² "How the EU's Flawed Artificial Intelligence Regulation Endangers the Social Safety Net: Questions and Answers" Human Rights Watch (November 10, 2021)
<https://www.hrw.org/news/2021/11/10/how-eus-flawed-artificial-intelligence-regulation-endangers-social-safety-net#_Does_the_regulation_prevent>.

¹¹³ Access Now, "Access Now's Submission to the European Commission's Adoption Consultation on the Artificial Intelligence Act" (2021) page 15-16
<<https://www.accessnow.org/wp-content/uploads/2021/08/Submission-to-the-European-Commissions-Consultation-on-the-Artificial-Intelligence-Act.pdf>>.

¹¹⁴ Nathalie A Smuha and others, "How the EU Can Achieve Legally Trustworthy AI: A Response to the European Commission's Proposal for an Artificial Intelligence Act" [2021] Social Science Research Network
<https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3899991>.

¹¹⁵ EDPS-EDPB Joint Opinion, p. 11.

¹¹⁶ Access Now, "Access Now's Submission to the European Commission's Adoption Consultation on the Artificial Intelligence Act" (2021) page 15-16

How the GDPR and EU AI Act Safeguard Personal Data and Prevent Misuse in Social Scoring.

Article 5 of the GDPR mandates that personal data processing be lawful, fair, and transparent¹¹⁷. These principles are vital for social scoring activities, ensuring individuals know how their data is used and preventing unfair treatment. Both the GDPR and Article 5(1)(c) of the EU AI Act¹¹⁸ emphasise lawful, fair, and transparent data processing to protect fundamental rights and build trust.

Personal data must be collected for specific, explicit, and legitimate purposes and not further processed incompatibly¹¹⁹. Using data collected for one purpose, like social media activity, for social scoring without consent breaches this principle¹²⁰. The Schrems II case (C-311/18) by the Court of Justice of the European Union highlights the importance of the purpose limitation principle in data protection law, which mandates that personal data collected for one purpose cannot be repurposed without explicit consent¹²¹. Both the GDPR and Article 5(c) of the EU AI Act require personal data to be collected for legitimate purposes and not misused. Article 5(c) of the EU AI Act prohibits AI systems from evaluating individuals over

<<https://www.accessnow.org/wp-content/uploads/2021/08/Submission-to-the-European-Commissions-Consultation-on-the-Artificial-Intelligence-Act.pdf>>.

¹¹⁷ Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data (General Data Protection Regulation) [2016] OJ L119/1, art 5.

¹¹⁸ “Article 5: Prohibited Artificial Intelligence Practices | EU Artificial Intelligence Act” Section 1(c) <<https://artificialintelligenceact.eu/article/5/>>.

¹¹⁹ Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data (General Data Protection Regulation) [2016] OJ L119/1, art 5(1)(b).

¹²⁰ Court of Justice of the European Union, 'Data Protection Commissioner v Facebook Ireland Ltd and Maximillian Schrems (Case C-311/18)' (2020).

¹²¹ *ibid.*

time unless for a legitimate purpose and without leading to unjust treatment, ensuring transparency and protecting against misuse¹²².

Article 5(1)(c) of the GDPR mandates data minimisation, ensuring personal data is adequate, relevant, and limited to necessary purposes¹²³. Both the GDPR and Article 5(c) of the EU AI Act emphasise data minimisation to prevent excessive or irrelevant data collection. This protects against overreach in data processing, ensuring personal data is handled responsibly and ethically, which enhances trust in AI within the European Union.

Articles 13 and 14 of the GDPR require data controllers to provide clear, accessible information about data processing¹²⁴. This includes details on social scoring activities, ensuring individuals understand how their data is used. Both the GDPR and Article 5(c) of the EU AI Act stress transparency to promote trust and accountability in data processing. Providing clear information protects individuals' rights and allows them to contest data processing if necessary.

Article 22 of the GDPR grants individuals the right to avoid decisions based solely on automated processing that significantly affects them¹²⁵. Social scoring leading to significant decisions must include human intervention and the opportunity to contest. Both the GDPR

¹²² “Article 5: Prohibited Artificial Intelligence Practices | EU Artificial Intelligence Act” Section 1(c) <<https://artificialintelligenceact.eu/article/5/>>.

¹²³ Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data (General Data Protection Regulation) [2016] OJ L119/1, art 5(1)(c).

¹²⁴ Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data (General Data Protection Regulation) [2016] OJ L119/1, arts 13 and 14.

¹²⁵ Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data (General Data Protection Regulation) [2016] OJ L119/1, art 22.

and Article 5,(1)(c) of the EU AI Act address concerns about automated decision-making, emphasising human oversight to protect individuals' rights.

Article 9 of the GDPR imposes strict requirements on processing sensitive personal data, like racial or ethnic origin, political opinions, or religious beliefs¹²⁶. Processing such data for social scoring is prohibited unless there is explicit consent¹²⁷ or another legal basis for the purposes of Article 9(2). Both the GDPR and Article 5,1(c) of the EU AI Act stress this heightened protection to safeguard sensitive information, ensuring it is not misused by AI systems for social scoring. This approach aligns with preserving fundamental rights and privacy.

Overall, the GDPR and the EU AI Act work together to ensure personal data is processed lawfully, fairly, and transparently, preventing misuse and protecting individuals' rights in social scoring contexts. These regulations foster trust and accountability in data processing, ensuring responsible and ethical handling of personal data.

AI Act's Targeted Approach vs. GDPR's Comprehensive Framework.

One significant difference between the EU AI Act's provision in Article 5(1)(c) and the GDPR is the specific focus on AI systems and their use for social scoring. While the GDPR provides a comprehensive framework for the protection of personal data, including rules on automated decision-making and profiling¹²⁸, Article 5(c) of the EU AI Act specifically addresses the evaluation or classification of individuals by AI systems based on social behavior or personal characteristics “over-time”¹²⁹. This targeted approach reflects growing concerns over the potential risks associated with AI use for social scoring and its impact on individuals' rights, freedoms, and societal well-being. The EUAI Act highlights the specific

¹²⁶ Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data (General Data Protection Regulation) [2016] OJ L119/1, art 9.

¹²⁷ Ibid Article 9(2)(a).

¹²⁸ European Union, 'General Data Protection Regulation (GDPR)' (2016), arts 22 and Recital 71.

¹²⁹ “Article 5: Prohibited Artificial Intelligence Practices | EU Artificial Intelligence Act” Section 1(c) <<https://artificialintelligenceact.eu/article/5/>>.

risks and challenges posed by AI systems in the context of social scoring, which may not be fully addressed by the broader data protection framework provided by the GDPR.

In contrast, the GDPR provides a broader framework for data protection and privacy, encompassing various aspects of personal data processing beyond just social scoring¹³⁰. It includes comprehensive provisions on automated decision-making¹³¹, profiling, data minimisation¹³², purpose limitation¹³³, and the protection of special categories of data. Therefore, while Article 5(1)(c) of the EU AI Act addresses specific issues related to AI systems and social scoring, compliance with the GDPR remains essential for all data processing activities, including those involving AI, to ensure comprehensive protection of individuals' rights and freedoms.

Chapter 4: Analysis of Biometric Identification and RTBI under the EU AI Act.

Regulating Biometric Identification and Real-Time Remote Biometric Systems Under the EU AI Act.

Biometric identification verifies identity using unique physical or behavioural traits, such as fingerprints, facial recognition, or iris scans¹³⁴. Real-time remote biometric identification employs surveillance cameras or sensors with AI to recognise individuals in public areas

¹³⁰ European Union, 'General Data Protection Regulation (GDPR)' (2016).

¹³¹ Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data (General Data Protection Regulation) [2016] OJ L119/1, art 22.

¹³² "Article 5: Prohibited Artificial Intelligence Practices | EU Artificial Intelligence Act" Section 1(c) <<https://artificialintelligenceact.eu/article/5/>>.

¹³³ "Article 5: Prohibited Artificial Intelligence Practices | EU Artificial Intelligence Act" Section 1(b) <<https://artificialintelligenceact.eu/article/5/>>.

¹³⁴K. Delac and M. Grgic, "A survey of biometric recognition methods," Proceedings. Elmar-2004. 46th International Symposium on Electronics in Marine, Zadar, Croatia, 2004, pp. 184-193.

instantly, often without their consent. This process raises significant privacy concerns and risks to fundamental rights due to the invasive nature of continuous surveillance.

The EU AI Act proposes to prohibit real-time remote biometric systems in public spaces for law enforcement unless strictly necessary, and it also bans AI from making risk assessments solely based on profiling¹³⁵. The regulation categorizes biometric systems into remote and traditional types to provide a clear framework for their use and ensure compliance with legal standards¹³⁶.

Biometric data refers to personal data resulting from specific technical processing related to the physical, physiological, or behavioural characteristics of an individual, which allows or confirms the unique identification of that individual¹³⁷. This includes but is not limited to, fingerprint data, facial recognition, and iris scans¹³⁸.

Real-time biometric identification (RTBI) involves the use of AI and biometric systems, such as surveillance cameras or sensors, to instantly recognise individuals in public spaces¹³⁹. This process often occurs without the individual's consent, raising significant privacy concerns.

¹³⁵ “Article 5: Prohibited Artificial Intelligence Practices | EU Artificial Intelligence Act” section 1(g) <<https://artificialintelligenceact.eu/article/5/>>.

¹³⁶ Nathalie A Smuha and others, “How the EU Can Achieve Legally Trustworthy AI: A Response to the European Commission’s Proposal for an Artificial Intelligence Act” [2021] Social Science Research Network <https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3899991>.

¹³⁷ Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data (General Data Protection Regulation) [2016] OJ L119/1, art 4(14).

¹³⁸ J Daugman, “How Iris Recognition Works” (2004) 14 IEEE Transactions on Circuits and Systems for Video Technology 21 <<https://doi.org/10.1109/tcsvt.2003.818350>>. vol. 14, no. 1, pp. 21-30.

¹³⁹ R Penfold, 'Review of Surveillance Society: Monitoring Everyday Life; Everyday Surveillance: Vigilance and Visibility in Post Modern Life, by D Lyon & W Staples' (2002) 42(1) The British Journal of Criminology 222-224 <http://www.jstor.org/stable/23638774>

RTBI systems typically operate continuously, identifying individuals as they move through public areas, thereby potentially impacting their fundamental rights and freedoms¹⁴⁰.

Enforcement Challenges in the EU AI Act's Biometric Provisions.

The definition of "biometric categorisation system" in Article 3(35)¹⁴¹ of the EU AI Act faces criticism for several shortcomings. It introduces ambiguity by allowing systems deemed "ancillary to another commercial service," yet it lacks clarity on what constitutes ancillary. This vagueness opens the door for potential misuse, as the requirement for strict necessity based on technical reasons is not well-defined¹⁴². Consequently, allowing biometric systems to operate as ancillary to other services may create loopholes that compromise privacy and rights without clear justification.

This provision aims to classify biometric systems but fails to provide precise boundaries, leading to concerns about the enforcement and practical application of the law¹⁴³. The lack of clear definitions could undermine the act's effectiveness in safeguarding individuals' privacy. Undermining the EU's mandate for trustworthy AI.

Article 5, paragraph 1(g) of the EU AI Act exempts law enforcement from regulations concerning the categorisation of data¹⁴⁴, raising significant concerns about potential misuse.

¹⁴⁰ CJ Hoofnagle, B Van Der Sloot and F Zuiderveen Borgesius, "The European Union General Data Protection Regulation: What It Is and What It Means" [2019] Information & Communications Technology Law 65
<https://pure.uva.nl/ws/files/34085945/European_Union_General_Data_Protection_Regulation.pdf>.

¹⁴¹ "Article 3: Definitions | EU Artificial Intelligence Act" section 3(35)
<<https://artificialintelligenceact.eu/article/3/>>.

¹⁴² Access Now, "Access Now's Submission to the European Commission's Adoption Consultation on the Artificial Intelligence Act" (2021) 16–17
<<https://www.accessnow.org/wp-content/uploads/2021/08/Submission-to-the-European-Commissions-Consultation-on-the-Artificial-Intelligence-Act.pdf>>.

¹⁴³ "Article 3: Definitions | EU Artificial Intelligence Act" section 3(35)
<<https://artificialintelligenceact.eu/article/3/>>.

¹⁴⁴ "Article 5: Prohibited Artificial Intelligence Practices | EU Artificial Intelligence Act" s 5 (g) <<https://artificialintelligenceact.eu/article/5/>>.

This exemption allows law enforcement agencies to deploy remote biometric identification systems (RBIS) under the guise of security, leading to a slippery slope where privacy rights could be compromised.

Such discretion can lead to function creep, where biometric technologies are applied beyond their original intent, significantly eroding public trust and infringing on civil liberties¹⁴⁵. The lack of stringent oversight or clear limitations poses serious privacy risks and increases the potential for discriminatory practices and unjustified surveillance¹⁴⁶. These issues highlight the need for more precise regulatory measures to protect fundamental rights and prevent abuse by law enforcement agencies.

The Ethical and Privacy Concerns of Facial Recognition Use in Protests and Law Enforcement.

During the Black Lives Matter protests in 2020, reports surfaced that law enforcement agencies in the United States were using facial recognition technology to identify and track protesters¹⁴⁷. This practice raises several concerns, including a chilling effect on free speech, as the use of facial recognition at protests can deter individuals from exercising their right to protest and freedom of assembly, knowing they might be identified and monitored¹⁴⁸.

¹⁴⁵ European Digital Rights (EDRi), “European Commission Adoption Consultation: Artificial Intelligence Act” (2021) <<https://edri.org/wp-content/uploads/2021/08/European-Digital-Rights-EDRi-submission-to-European-Commission-adoption-consultation-on-the-Artificial-Intelligence-Act-August-2021.pdf>>.

¹⁴⁶ European Union Agency for Fundamental Rights, “Facial Recognition Technology: Fundamental Rights Considerations in the Context of Law Enforcement” (2019) <https://fra.europa.eu/sites/default/files/fra_uploads/fra-2019-facial-recognition-technology-focus-paper-1_en.pdf>.

¹⁴⁷ Amnesty International, “USA: NYPD Ordered to Hand over Documents Detailing Surveillance of Black Lives Matter Protests Following Lawsuit” (Amnesty International, August 4, 2022) <<https://www.amnesty.org/en/latest/news/2022/08/usa-nypd-black-lives-matter-protests-surveillance/#:~:text=The%20New%20York%20Police%20Department,the%20Surveillance%20Technology%20Oversight%20Project>>.

¹⁴⁸ “The Fight to Stop Face Recognition Technology | American Civil Liberties Union” (American Civil Liberties Union, June 7, 2023) <<https://www.aclu.org/news/topic/stopping->

Furthermore, there are concerns about the disproportionate targeting of certain groups, particularly racial minorities, which could lead to further discrimination and bias in law enforcement practices¹⁴⁹. Additionally, there is often a lack of accountability regarding how and why such technology is used, raising the risk of abuse.

These issues highlight significant violations of fundamental rights, such as the right to freedom of expression¹⁵⁰ and the right to freedom of assembly and association¹⁵¹. The unchecked use of Real-time Biometric Identification Systems (RBIS) by law enforcement agencies raises ethical, privacy, and human rights concerns. Despite their potential benefits in enhancing security¹⁵², these technologies can lead to negative consequences if used without proper oversight. For instance, Biometric Categorisation Systems (BCS) can exacerbate discriminatory practices by categorising individuals based on sensitive characteristics like race and gender, leading to biased profiling. Emotion Recognition Systems (ERS) are also problematic as they lack scientific reliability and can be intrusive, claiming to infer emotions from physiological signals without consistent accuracy¹⁵³.

Moreover, the pervasive surveillance enabled by RBIS infringes on individuals' privacy rights, as continuous monitoring and tracking without consent constitute significant privacy

face-recognition-

surveillance#:~:text=Face%20recognition%20surveillance%20presents%20an,places%20of%20worship%2C%20and%20more>.

¹⁴⁹ Patrick Grother and others, 'Face Recognition Vendor Test (FRVT) Part 3: Demographic Effects' (National Institute of Standards and Technology 2019)

<https://nvlpubs.nist.gov/nistpubs/ir/2019/NIST.IR.8280.pdf> accessed 22 May 2024.

¹⁵⁰ Charter of Fundamental Rights of the European Union [2012] OJ C326/391, art 11.

¹⁵¹ Charter of Fundamental Rights of the European Union [2012] OJ C326/391, art 12.

¹⁵² Aware, "Facial Recognition & Law Enforcement – the Value Proposition" (Aware, May 15, 2024) <<https://www.aware.com/blog-facial-recognition-used-in-law-enforcement/>>.

¹⁵³ Evan Selinger, "A.I. Can't Detect Our Emotions - OneZero" Medium (January 7, 2022) <<https://onezero.medium.com/a-i-cant-detect-our-emotions-3c1f6fce2539>>.

violations¹⁵⁴. These concerns underscore the need for stringent regulatory measures to limit law enforcement's use of biometric identification systems and protect individuals' fundamental rights.

Rationale Behind the Prohibition The Regulation of Real-Time Biometric Identification Systems Under the EU AI Act.

Real-time biometric identification (RTBI) systems involve the continuous collection, processing, and analysis of biometric data, such as facial images, without the explicit consent of individuals¹⁵⁵. This pervasive surveillance poses significant risks to individuals' privacy. The European Commission's proposal for the Artificial Intelligence Act highlights these concerns, aiming to protect privacy and personal data as enshrined in Articles 7 and 8 of the EU Charter of Fundamental Rights by prohibiting RTBI systems in public spaces¹⁵⁶. For instance, preventing the constant tracking and identification of individuals in public spaces ensures that their daily activities and movements are not subject to unwarranted scrutiny, a stance supported by the European Data Protection Supervisor (EDPS) in their critique of facial recognition technologies¹⁵⁷.

¹⁵⁴“The Dawn of Robot Surveillance | American Civil Liberties Union” (American Civil Liberties Union, June 11, 2019) <<https://www.aclu.org/publications/dawn-robot-surveillance>>.

¹⁵⁵ Publications Office of the European Union, “Study to Support an Impact Assessment of Regulatory Requirements for Artificial Intelligence in Europe : Final Report.” (Publications Office of the EU, 2021) Recital 27 <<https://op.europa.eu/en/publication-detail/-/publication/55538b70-a638-11eb-9585-01aa75ed71a1/language-en>>.

¹⁵⁶ “EUR-LEX - 12012P/TXT - EN - EUR-LEX” <<https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:12012P/TXT>>.

¹⁵⁷ “Facial Recognition: A Solution in Search of a Problem?” (European Data Protection Supervisor, October 28, 2019) <https://www.edps.europa.eu/press-publications/press-news/blog/facial-recognition-solution-search-problem_en>.

The unrestricted use of real-time biometric identification (RTBI) systems could lead to the development of a surveillance infrastructure¹⁵⁸, eroding public trust in the state and its institutions, thereby undermining the foundational values of democracy and transparency. To prevent the emergence of a surveillance state, Article 5 of the EU AI Act imposes strict limitations on the use of biometric identification technologies. Prohibiting mass surveillance practices helps maintain a balance between state security needs and individual freedoms, fostering trust in governmental institutions¹⁵⁹. By restricting the use of RTBI systems, the regulation ensures that state surveillance does not overreach, thereby protecting individual privacy and supporting democratic principles.

The use of real-time biometric identification (RTBI) systems at protests and public gatherings can have a chilling effect on individuals' willingness to exercise their rights to freedom of assembly and expression. The fear of being identified and tracked can deter people from participating in peaceful demonstrations and voicing dissent¹⁶⁰. Article 5's prohibition helps safeguard these fundamental democratic rights, ensuring that individuals can freely express their opinions and assemble without fear of surveillance¹⁶¹. By ensuring that protestors are not subject to real-time surveillance, the regulation supports robust democratic participation and protects civic engagement¹⁶². This approach aligns with the need to maintain an open and

¹⁵⁸ “Biometrics | Privacy International”

<[¹⁵⁹ Jennifer Lynch and others, “Face Off” \(Gennie Gebhart ed, 2019\)](https://privacyinternational.org/learn/biometrics#:~:text=What%20is%20the%20problem,discrimination%2C%20profiling%20and%20mass%20surveillance.>.</p></div><div data-bbox=)

<[>.](https://www.eff.org/files/2019/05/28/face-off-report.pdf)

¹⁶⁰ Anna Bacciarelli, “Time to Ban Facial Recognition from Public Spaces and Borders”

Human Rights Watch (September 29, 2023) <[>.](https://www.hrw.org/news/2023/09/29/time-ban-facial-recognition-public-spaces-and-borders)

¹⁶¹ Amnesty International, “Ban the Scan - Amnesty International” (Amnesty International,

February 11, 2022) <[¹⁶² “The Fight to Stop Face Recognition Technology | American Civil Liberties Union”](https://www.amnesty.org/en/petition/ban-the-scan-petition/#:~:text=World%20governments%20must%20acknowledge%20and,feel%20safer%20in%20their%20communities.>.</p></div><div data-bbox=)

(American Civil Liberties Union, June 7, 2023) <[>.](https://www.aclu.org/news/topic/stopping-face-recognition-surveillance)

free society where individuals can engage in civic activities without undue fear of government monitoring.

Biometric identification technology offers convenience but raises significant ethical and legal concerns about privacy, discrimination, and accountability¹⁶³. The lack of scientific validation for emotion recognition and sentiment analysis undermines their reliability, especially in contexts like hiring. Facial recognition systems have high error rates and biases, prompting calls for regulatory intervention. Remote biometric identification (RBI) in public spaces infringes on individual rights and democracy, necessitating strict regulatory measures and oversight. The EU AI Act addresses these issues by regulating the use of biometric technologies to protect privacy and ensure ethical AI deployment.

Alignment with EU Legal Frameworks: Safeguarding Privacy and Law Enforcement Needs.

Article 5 of the EU AI Act provides exceptions to the prohibition on real-time remote biometric identification systems for law enforcement purposes¹⁶⁴. These exceptions allow for the use of such systems in targeted searches for specific potential crime victims, the prevention of specific, substantial, and imminent threats to life or physical safety, and the detection, localisation, identification, or prosecution of perpetrators or suspects of criminal offenses punishable by custodial sentences or detention orders of at least three years, as determined by relevant Member State law. These provisions align with Article 2(2) of Council Framework Decision 2002/584/JHA¹⁶⁵, which outlines the serious offenses subject to the European Arrest Warrant and extradition procedures. By incorporating these exceptions, the EU AI Act ensures that while prioritising individual privacy and preventing misuse, it also provides necessary tools for law enforcement to address significant threats and

¹⁶³ Publications Office of the European Union, “Study to Support an Impact Assessment of Regulatory Requirements for Artificial Intelligence in Europe : Final Report.” (Publications Office of the EU, 2021) <<https://op.europa.eu/en/publication-detail/-/publication/55538b70-a638-11eb-9585-01aa75ed71a1/language-en>>.

¹⁶⁴ “Article 5: Prohibited Artificial Intelligence Practices | EU Artificial Intelligence Act” s 5 (g) <<https://artificialintelligenceact.eu/article/5/>>.

¹⁶⁵ Council Framework Decision 2002/584/JHA of 13 June 2002 on the European arrest warrant and the surrender procedures between Member States [2002] OJ L190/1, art 2(2).

serious crimes. This balance underscores the EU's commitment to upholding democratic values, ensuring that security measures are judiciously applied to protect both public safety and individual freedoms¹⁶⁶.

Both the GDPR and the EU AI Act emphasise the need for lawful, fair, and transparent processing of personal data. The EU AI Act's provisions on real-time biometric identification (RTBI) ensure that the deployment of biometric identification technologies is transparent, with clear guidelines and public oversight. This is in line with GDPR Article 5(1)(a), which states that personal data must be processed lawfully, fairly, and in a transparent manner¹⁶⁷. Additionally, the EU AI Act aligns with the GDPR's purpose limitation principle by restricting the use of biometric data to specific, well-defined purposes, preventing misuse of data collected for one purpose from being used arbitrarily for another. This is consistent with GDPR Article 5(1)(b), which mandates that data must be collected for specified, explicit, and legitimate purposes and not further processed in a manner incompatible with those purposes¹⁶⁸. These provisions ensure that both regulations work together to protect individual privacy and maintain trust in the use of AI and data processing technologies.

The EU AI Act aligns with the Law Enforcement Directive (Directive (EU) 2016/680)¹⁶⁹ by setting stringent conditions for the processing of biometric data for law enforcement purposes. This alignment ensures that such processing is necessary, proportionate, and subject

¹⁶⁶ Consolidated Version of the Treaty on European Union [2012] OJ C326/13.

¹⁶⁷ Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data (General Data Protection Regulation) [2016] OJ L119/1, art 5(1)(a).

¹⁶⁸ GDPR Article 5(1)(a): Personal data must be processed lawfully, fairly, and in a transparent manner.

¹⁶⁹ Directive (EU) 2016/680 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data by competent authorities for the purposes of the prevention, investigation, detection or prosecution of criminal offences or the execution of criminal penalties, and on the free movement of such data (Law Enforcement Directive) [2016] OJ L119/89, arts 8 and 10.

to oversight. The Law Enforcement Directive, particularly Articles 8¹⁷⁰ and 10¹⁷¹, mandates that personal data must be processed lawfully and fairly, and that data processing for law enforcement purposes must be strictly necessary and proportionate. Similarly, the EU AI Act imposes strict limitations and oversight on the use of real-time biometric identification systems by law enforcement, ensuring these technologies are deployed in a manner that protects fundamental rights and maintains public trust.

These provisions ensure that the deployment of biometric identification technologies adheres to high standards of legality, necessity, and proportionality, reinforcing the principles laid out in the Law Enforcement Directive. By doing so, the EU AI Act ensures that biometric data used for law enforcement purposes is handled with the utmost care, respecting individuals' privacy and upholding the rule of law within the EU.

Impact on the Rule of Law.

AI systems in law enforcement, public administration, and the judiciary can undermine accountability and transparency, leading to public distrust and unfair legal outcomes. The lack of explainability in AI decision-making can also embed errors or biases within legal rules, threatening the broader societal interest in a transparent and accountable legal system¹⁷².

¹⁷⁰ Directive (EU) 2016/680 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data by competent authorities for the purposes of the prevention, investigation, detection or prosecution of criminal offences or the execution of criminal penalties, and on the free movement of such data (Law Enforcement Directive) [2016] OJ L119/89, art 8.

¹⁷¹ Directive (EU) 2016/680 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data by competent authorities for the purposes of the prevention, investigation, detection or prosecution of criminal offences or the execution of criminal penalties, and on the free movement of such data (Law Enforcement Directive) [2016] OJ L119/89, art 10.

¹⁷² Nathalie A Smuha, “Beyond the Individual: Governing AI’s Societal Harm” (Internet Policy Review, October 13, 2021) <<https://policyreview.info/articles/analysis/beyond-individual-governing-ais-societal-harm>>.

Chapter 5: EUAI Office

The Pivotal Role of the European AI Office in AI Governance.

The European AI Office is pivotal in supporting the development and use of trustworthy AI while mitigating associated risks, laying the foundation for a unified European AI governance system¹⁷³. In light of the rapid advancement of AI technologies and the European Union's ambitious goal to substantially increase AI adoption among businesses, regulatory measures are essential for ensuring responsible AI development and deployment. The European AI Office serves as the central authority overseeing AI developments, enforcing regulations, and fostering international cooperation¹⁷⁴. The European Commission's target for 75% of European businesses to utilise AI by the end of the decade highlights the transformative potential of this technology,¹⁷⁵ promising significant impacts on economic growth, innovation, and competitiveness. However, without appropriate regulations, the unchecked deployment of AI can lead to risks such as bias, discrimination, and privacy violations. Therefore, ethical considerations must guide AI development, with regulations playing a crucial role in promoting responsible practices.

The definition of the "AI Office"¹⁷⁶ clarifies that it is the Commission's function responsible for implementing, monitor, and supervising AI systems and general-purpose AI models, as well as AI governance. This aligns with the mandate and responsibilities of investigating possible infringements or rules and ensuring compliance, as well as, providing advice , guidance and support structures for AI uptake.

¹⁷³ “European AI Office: Shaping Trustworthy AI in Europe” Digital Skills and Jobs Platform (March 21, 2024) <<https://digital-skills-jobs.europa.eu/en/latest/news/european-ai-office-shaping-trustworthy-ai-europe>>.

¹⁷⁴ *ibid*

¹⁷⁵ Benjamin Mueller, “How Much Will the Artificial Intelligence Act Cost Europe?” (Center for Data Innovation, July 26, 2021) <<https://datainnovation.org/2021/07/how-much-will-the-artificial-intelligence-act-cost-europe/>>.

¹⁷⁶ “Article 3: Definitions | EU Artificial Intelligence Act” s 3(47) <<https://artificialintelligenceact.eu/article/3/>>.

The European AI Office, as empowered by the AI Act, holds a significant mandate and authority to ensure the coherent application of AI regulations across Member States¹⁷⁷. This office supports the AI Act by enforcing rules related to general-purpose AI, developing tools, methodologies, and benchmarks for evaluating AI models, and drawing up codes of practice in cooperation with stakeholders¹⁷⁸. It also investigates possible infringements of AI regulations and ensures compliance, strengthening the development and use of trustworthy AI across the EU¹⁷⁹. Additionally, the office provides advice, guidance, and support structures for AI uptake, fostering international cooperation and governance on AI¹⁸⁰. These responsibilities are outlined in various policy documents and initiatives by the European Commission, emphasising the comprehensive role of the European AI Office in the AI regulatory landscape.

National Efforts and Regulatory Challenges in AI Act Implementation.

Spain demonstrated a strong commitment to AI regulation by proactively setting up the Agency for the Supervision of Artificial Intelligence (AESIA) in A Coruña in 2023¹⁸¹. The Netherlands followed by establishing a department within its data protection authority to deal with algorithms, currently staffed by 12 employees with plans to expand to 20¹⁸², showing a growing but still limited capacity for handling AI regulation. Ireland's Department of Enterprise, Trade, and Employment is leading the development of the national

¹⁷⁷ “European AI Office: Shaping Trustworthy AI in Europe” Digital Skills and Jobs Platform (March 21, 2024) <<https://digital-skills-jobs.europa.eu/en/latest/news/european-ai-office-shaping-trustworthy-ai-europe>>.

¹⁷⁸ “European AI Office” (Shaping Europe’s Digital Future) <<https://digital-strategy.ec.europa.eu/en/policies/ai-office>>.

¹⁷⁹ *ibid.*

¹⁸⁰ *ibid.*

¹⁸¹ Cynthia Kroet, “As the EU AI Act Enters into Force, Focus Shifts to Countries’ Oversight Appointments” Euronews (March 12, 2024) <<https://www.euronews.com/next/2024/03/12/as-ai-act-enters-into-force-focus-shifts-to-countries-oversight-appointments>>.

¹⁸² *Ibid.*

implementation plan for the AI Act, though there is hesitancy to speculate on enforcement arrangements until the EU legislative process is completed. Luxembourg is actively consulting stakeholders to implement the AI Act¹⁸³, emphasising the need for efficient coordination among regulators. Despite these efforts, there is widespread concern about the effective implementation and enforcement of the AI Act¹⁸⁴. As the rules are on the verge of entering into force, the lack of clear national arrangements for enforcement, as noted by Ireland, indicates potential gaps in regulatory oversight.

Member States are granted authority to designate or establish national competent authorities for enforcement, but concerns persist regarding uneven implementation and resource disparities¹⁸⁵. Broad exceptions for Member States, including those for public security reasons, pose risks to fundamental rights protection. Provisions granting supervisory authorities access to data and documentation must balance regulatory needs with the protection of intellectual property. Additionally, the reliance on conformity assessments and the absence of individual redress mechanisms are identified as shortcomings in the enforcement structure¹⁸⁶. These issues underscore the critical role of the European AI Office and the European AI Board in providing oversight and ensuring harmonized implementation across Member States. The European AI Board is tasked with coordinating efforts and addressing disparities in enforcement practices, while the European AI Office supports Member States by providing guidance and fostering cooperation among national regulators. This centralised approach aims to mitigate the risks associated with national disparities and exceptions, enhancing the overall effectiveness of the AI Act's regulatory framework.

Ensuring Adequate Resources and Addressing Regulatory Challenges.

The adequacy of the European AI Office's resources is ensured through a hybrid financing approach that combines internal resource reallocation and specific budgetary allocations. Staffing needs are primarily met through personnel from the Directorate-General. However,

¹⁸³ Ibid.

¹⁸⁴ Ibid.

¹⁸⁵ Proposal for a Regulation of the European Parliament and of the Council laying down harmonized rules on artificial intelligence (Artificial Intelligence Act)," Article 6, European Commission, 2021

¹⁸⁶ "EU AI Act Compliance Checker | EU Artificial Intelligence Act"

<<https://artificialintelligenceact.eu/assessment/eu-ai-act-compliance-checker/>>.

concerns have been raised about the staffing levels required to enforce the Draft AI Act at the Member State level. The estimate that enforcement will require between 1 and 25 extra full-time staff per Member State is considered dangerously optimistic¹⁸⁷, suggesting a potential underestimation of the resources needed for effective implementation and oversight.

Regulatory challenges persist in updating regulations promptly to keep pace with technological advancements, as evidenced by historical experiences across various arms control regimes¹⁸⁸. Similarly, past struggles highlight the difficulty in regulatory adaptation as technology evolves¹⁸⁹.

Conclusion.

The EU AI Act represents a commendable effort towards establishing a comprehensive regulatory framework for artificial intelligence within Europe. The EU AI Act's emphasis on defending fundamental rights is critical in an era where AI technologies are rapidly influencing personal and social domains¹⁹⁰. By enshrining values that prioritise human dignity, privacy, and autonomy, the Act seeks to reduce the dangers associated with AI systems that would otherwise violate these fundamental rights.

¹⁸⁷ Michael Veale and Frederik J Zuiderveen Borgesius, “Demystifying the Draft EU Artificial Intelligence Act — Analysing the Good, the Bad, and the Unclear Elements of the Proposed Approach” (2021) 22 Computer Law Review International 97 <<https://doi.org/10.9785/crl-2021-220402>>.

¹⁸⁸ Nelson, Amy J. and Nelson, Amy J., Innovation Acceleration, Digitization, and the Arms Control Imperative (March 26, 2019). Available at SSRN: <https://ssrn.com/abstract=3382956>

¹⁸⁹ Picker, Colin, A View from 40,000 Feet: International Law and the Invisible Hand of Technology. Cardozo Law Review, Vol. 232, p. 149, 2001, Available at SSRN: <https://ssrn.com/abstract=987524>

¹⁹⁰ “EUR-LEX - 52021PC0206 - EN - EUR-LEX” <<https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:52021PC0206>>.

The Act prevents AI systems from manipulating human behavior, protecting privacy and autonomy. Transparency is a key component of the EU AI Act¹⁹¹, ensuring that AI activities are explicit and intelligible to users. This emphasis on openness is crucial to establishing confidence between AI developers and consumers. Transparent AI solutions allow consumers to understand how their data is handled and the reasoning behind AI-driven choices. This is crucial for preventing AI systems from perpetuating prejudices and discrimination without user knowledge. The Act's provisions for ethical AI practices emphasise the significance of creating and deploying AI in ways that are consistent with society values and ethical norms. The Act prohibits unethical use of AI, including social scoring systems that unfairly rate persons based on behavior or attributes, to avoid discrimination and social inequity¹⁹².

One of the Act's key merits is its restriction on manipulative AI approaches¹⁹³. By prohibiting AI systems that cause harm, the Act safeguards people from hidden effects that may damage their decision-making processes or undermine other fundamental human rights. This law protects user autonomy and ensures ethical and transparent usage of AI technologies. The Act also aims to prohibit the abuse of vulnerable groups. By prohibiting AI systems from targeting individuals based on age, handicap, or socioeconomic position, the Act assures that AI technologies do not worsen existing inequities or exploit those who are less able to defend themselves¹⁹⁴. This method supports the objective of promoting social justice and safeguarding persons from dangerous AI activities. Another important component of the EU AI Act is its limitation on social scoring systems. These systems, which assess individuals

¹⁹¹ “Ethics Guidelines for Trustworthy AI” (Shaping Europe’s Digital Future, April 8, 2019) <<https://digital-strategy.ec.europa.eu/en/library/ethics-guidelines-trustworthy-ai>>.

¹⁹² Yeung, Karen and Howes, Andrew and Pogrebna, Ganna, AI Governance by Human Rights-Centred Design, Deliberation and Oversight: An End to Ethics Washing (June 21, 2019). Forthcoming in M Dubber and F Pasquale (eds.) *The Oxford Handbook of AI Ethics*, Oxford University Press (2019) , Available at SSRN: <https://ssrn.com/abstract=3435011>

¹⁹³ Sandra Wachter, Brent Mittelstadt and Luciano Floridi, “Why a Right to Explanation of Automated Decision-Making Does Not Exist in the General Data Protection Regulation” (2017) 7 *International Data Privacy Law* 76 <<https://doi.org/10.1093/idpl/ix005>>.

¹⁹⁴ Virginia Eubanks, *Automating Inequality: How High-Tech Tools Profile, Police, and Punish the Poor* (St Martin’s Press 2018).

based on their behavior or attributes, pose substantial threats to privacy and fairness. By prohibiting such methods, the Act bans the creation of monitoring systems that might lead to unfair discrimination and societal control, comparable to China's Social Credit System. This restriction demonstrates the EU's commitment to protecting democratic norms and individual liberties.

Despite these advantages, there are still areas where the EU AI Act might be enhanced to increase its efficacy and comprehensiveness. One critical area for development is the need for more precise definitions and tighter enforcement methods¹⁹⁵. Ambiguous wording like "materially distort a person's behavior" and "beyond a person's consciousness" can make enforcement difficult and open up opportunities for ethical problems and loopholes¹⁹⁶. Refining these definitions and offering more precise recommendations will assist to guarantee that AI systems do not use legislative ambiguity to manipulate or harm persons.

Moreover, the Act's scope could be expanded to address all forms of manipulation, not just subliminal techniques¹⁹⁷. Explicit manipulative practices that are not strictly subliminal but still harmful need to be regulated to provide comprehensive protection against various manipulative technologies. Additionally, the Act should broaden its definition of harm¹⁹⁸ to include economic, social, and cultural harms, ensuring a more thorough protection framework that aligns with the evolving nature of AI technologies.

The alignment of the EU AI Act with existing legal frameworks such as the GDPR and the Law Enforcement Directive is crucial for creating a cohesive regulatory environment. Ensuring that these regulations work together to protect individual privacy, prevent misuse of

¹⁹⁵ Michael Veale and Frederik J Zuiderveen Borgesius, "Demystifying the Draft EU Artificial Intelligence Act — Analysing the Good, the Bad, and the Unclear Elements of the Proposed Approach" (2021) 22 Computer Law Review International 97 <<https://doi.org/10.9785/crl-2021-220402>>.

¹⁹⁶ Access Now, "Access Now's Submission to the European Commission's Adoption Consultation on the Artificial Intelligence Act" (2021) <<https://www.accessnow.org/wp-content/uploads/2021/08/Submission-to-the-European-Commissions-Consultation-on-the-Artificial-Intelligence-Act.pdf>>.

¹⁹⁷ Ibid.

¹⁹⁸ Ibid.

personal data, and maintain public trust is essential for fostering an ethical and responsible AI landscape. Enforcement of GDPR required significant investment in regulatory bodies, training, and public awareness campaigns. Governments had to ensure that the regulations were not only enforceable but also understood by the businesses required to comply¹⁹⁹.

Arguably this would be even more prominent in the adoption of the EUAI Act because of the complex nature of AI and the lack of public understanding surrounding it.

To summarise, while the EU AI Act represents a substantial step forward in AI legislation, more improvements are required to solve its existing weaknesses. The EU should better protect basic rights and foster ethical AI development by revising definitions, broadening the scope of protections, and providing strong enforcement measures. These enhancements will assist to retain public trust in AI technology and support the EU's commitment to responsible and ethical AI innovation.

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¹⁹⁹ Information Commissioner’s Office, “Data Protection GDPR One Year On” (2019)
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